DTC P2104:00 [SKYACTIV-D 1.5]

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DTC P2104:00	Intake shutter valve control system: Idle air control
DETECTION CONDITION	• When all of the following conditions are met, the PCM determines that the exhaust gas pressure or boost pressure is the estimated value or more. — Ignition switched ON (engine on). — Shift position: except P or N position (ATX) — Shift lever position is neutral (MTX)
FAIL-SAFE FUNCTION	• Prevents a combustion pressure increase by closing the intake shutter valve.
POSSIBLE CAUSE	• Vacuum piping malfunction — Power brake unit—Vacuum pump — Vacuum pump—Check valve — Check valve—Vacuum chamber — Vacuum chamber—Turbocharger solenoid valve — Turbocharger solenoid valve—Turbocharger actuator • Oil suction into combustion chamber • Fuel injector No.1 malfunction • Fuel injector No.2 malfunction • Fuel injector No.3 malfunction • Fuel injector No.4 malfunction • Turbocharger with variable turbine geometry malfunction • Actuator position sensor malfunction • Turbocharger solenoid valve malfunction • Exhaust gas pressure sensor No.1 malfunction • MAP sensor malfunction • Vacuum chamber (built-into cylinder head cover) malfunction • Check valve malfunction • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	• Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**DETERMINE IF MALFUNCTION IS CAUSED BY TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY** • Switch the ignition off. • Switch the ignition ON (engine on). • Display PID VGTURB_ACT and simulation item VGTURB_DSD using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Operate the turbocharger with variable geometry using simulation item VGTURB_DSD. • Does PID VGTURB_ACT change according to the simulation operation?	Yes	Go to Step 12.
		No	Go to the next step.
5	**INSPECT VACUUM PIPING** • Inspect vacuum piping at the following: (See INTAKE-AIR SYSTEM VACUUM HOSE ROUTING DIAGRAM [SKYACTIV-D 1.5].) (See CONVENTIONAL BRAKE SYSTEM LOCATION INDEX.) — Power brake unit—Vacuum pump — Vacuum pump—Check valve — Check valve—Vacuum chamber — Vacuum chamber—Turbocharger solenoid valve — Turbocharger solenoid valve—Turbocharger actuator • Is there hose leakage or damage in the vacuum piping?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 9.
		No	Go to the next step.
6	**INSPECT VACUUM CHAMBER** • Inspect the vacuum chamber. (See VACUUM CHAMBER INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the cylinder head cover, then go to Step 9. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
7	**INSPECT TURBOCHARGER SOLENOID VALVE** • Inspect the turbocharger solenoid valve. (See TURBOCHARGER SOLENOID VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the turbocharger solenoid valve, then go to Step 9. (See TURBOCHARGER SOLENOID VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
8	**INSPECT CHECK VALVE** • Inspect the check valve. (See CHECK VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the check valve, then go to the next step. (See CHECK VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Replace the turbocharger with variable turbine geometry, then go to Step 12. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
9	**DETERMINE IF MALFUNCTION IS CAUSED BY TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY** • Switch the ignition off. • Switch the ignition ON (engine on). • Display PID VGTURB_ACT and simulation item VGTURB_DSD using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Operate the turbocharger with variable geometry using simulation item VGTURB_DSD. • Does PID VGTURB_ACT change according to the simulation operation?	Yes	Go to the next step.
		No	Replace the turbocharger with variable turbine geometry, then go to the next step. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

STEP	INSPECTION		ACTION
10	**INSPECT MAP SENSOR** • Reconnect all disconnected connectors. • Inspect the MAP sensor. (See MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the MAP sensor/boost air temperature sensor, then go to the next step. (See MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR/BOOST AIR TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
11	**INSPECT EXHAUST GAS PRESSURE SENSOR NO.1** • Reconnect all disconnected connectors. • Inspect the exhaust gas pressure sensor No.1. (See EXHAUST GAS PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the exhaust gas pressure sensor No.1, then go to the next step. (See EXHAUST GAS PRESSURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
12	**PERFORM COMPULSORY DPF REGENERATION AND BURN OUT ENGINE OIL IN DPF** • Perform the “COMPULSORY DIESEL PARTICULATE FILTER REGENERATION”. (See COMPULSORY DIESEL PARTICULATE FILTER REGENERATION [SKYACTIV-D 1.5].) **Note** • When a DTC is stored, compulsory DPF regeneration may be performed by the PCM control. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — For DTCs P242F:00, the "COMPULSORY DIESEL PARTICULATE FILTER REGENERATION" will only be blocked if an excessive amount of soot is in the diesel particulate filter. • Compulsory DPF regeneration is performed after performing the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].)	—	Go to the next step.
13	**INSPECT ENGINE OIL LEVEL** • Inspect the engine oil level. (See VACUUM CHAMBER INSPECTION [SKYACTIV-D 1.5].) • Is the engine oil amount above X line on the dipstick?	Yes	Replace the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) Perform the “ENGINE OIL DATA RESET”. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
14	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Perform the “FUEL INJECTOR INJECTION AMOUNT CORRECTION”. (See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].) • Start the engine and warm it up completely. **Caution** **• While performing this step, always operate the vehicle in a safe and lawful manner.** **• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.** • Drive the vehicle under the FREEZE FRAME DATA/snapshot data condition. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
15	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Referring to each adjustment procedure, perform fuel injector injection amount correction and then the diesel particulate filter data reset. (See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].) (See COMPULSORY DIESEL PARTICULATE FILTER REGENERATION [SKYACTIV-D 1.5].) DTC troubleshooting completed.